• Optimization - 3 (towards Gradient Descent)

Agenda ⇒

- ① Derivatives for optimization → Minima / Maxima
  - → Multivariate Calculus
  - → Partial Derivatives
  - → Gradients ⇒

Detailed Recap.

1] Derivatives: $f(x)$ → continuous & Differentiable at (a)

$$\frac{d}{dx}f(x) = f'(x) = \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$$

$\Delta x \to 0 \Rightarrow 0.0001$

|x|

Continuous ⇒ Differentiable

3] Rules:

(a) linearity → if $h(x) = a \cdot f(x) + b \cdot g(x)$ then $h'(x) = a f'(x) + b g'(x)$

(b) Product → if $h(x) = f(x) \cdot g(x)$ then $h'(x) = f'(x)g(x) + g'(x)f(x)$

(c) Quotient → if $h(x) = \frac{f(x)}{g(x)}$ then $h'(x) = \frac{g(x) \cdot f'(x) - g'(x) f(x)}{(g(x))^2}$

(d) Chain → if $h(x) = f(g(x))$ then $h'(x) = f'(g(x)) \cdot g'(x)$

$h(x) = f(x) \cdot \frac{1}{g(x)}$ ⇒ Product Rule

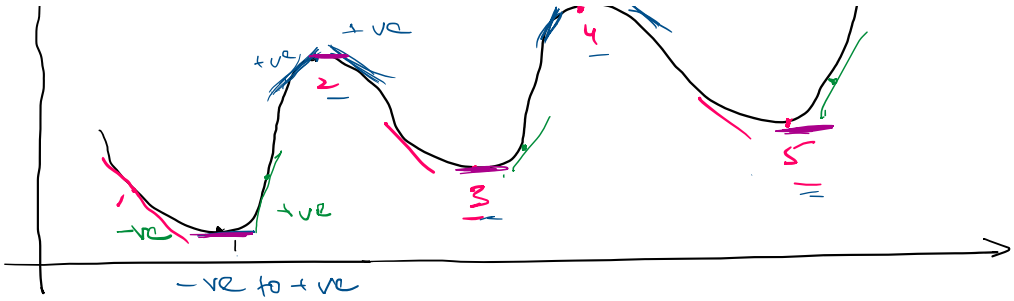
ML ⇒ $h(x) = f(g(x))$

$$h'(x) = f'(g(x)) \cdot g'(x)$$

⋆ How to use Derivatives for optimization ⇒

+ve or -ve slope



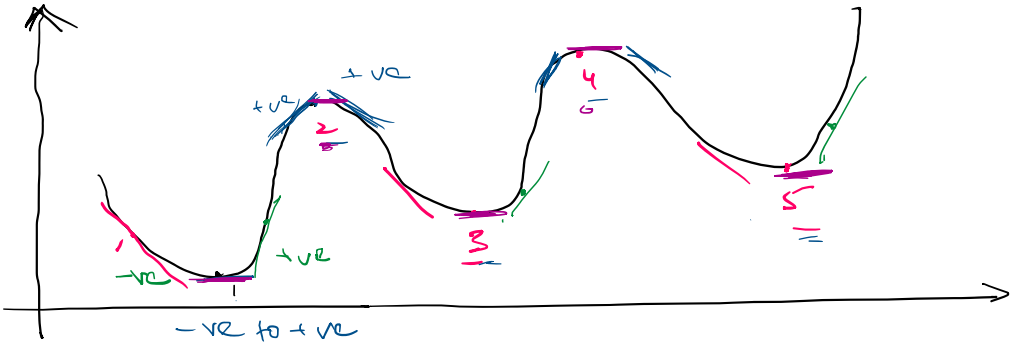


5 points → CRITICAL POINTS

1, 3, 5 → Minima

2, 4 → Maxima

How to identify critical points for any func?



slope at 1, 2, 3, 4, 5 → 0

1, 3, 5 → 1 ⇒ global minima

↓ ↳ Local minima ⇒ slope

(1) → global minima ✓

(4) → global maxima

(2) → local maxima

⇒ min (Local min) → global

max (Local max) → global

(2) → local maxima

↓ global

(*) Critical points ⇒

$f(x) \Rightarrow f'(x) = 0$

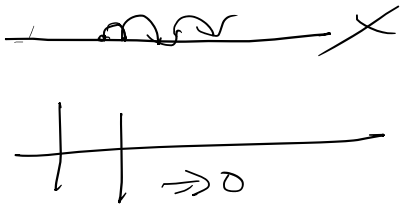
↳ 0 ⇒

maxima or minima

$f(x) = x^2 \Rightarrow f'(x) = 0$

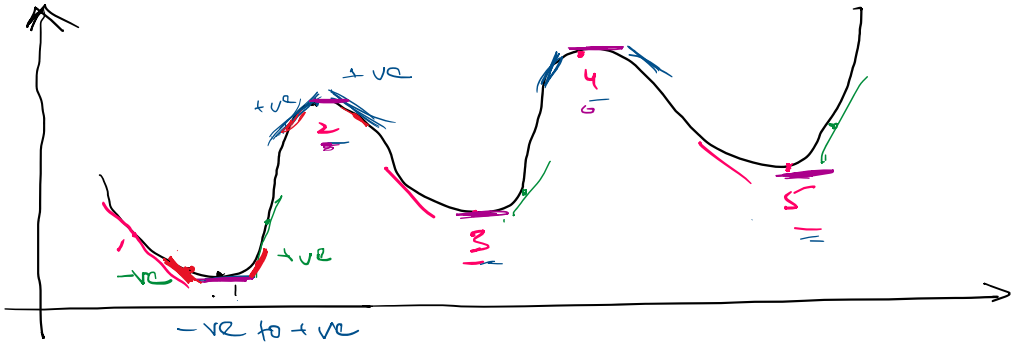
$2x = 0 \Rightarrow x = 0$ (*)

f(x) ⇒



(2) ⇒ 0

(*) How to identify maxima or minima?



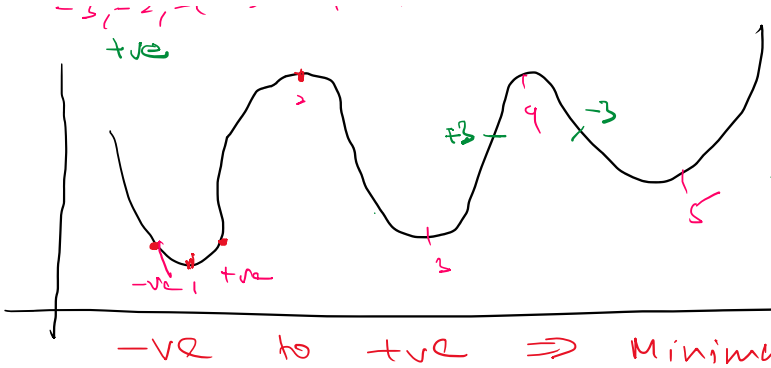
-ve to +ve ⇒ minima

+ve to -ve ⇒ maxima

-3, -2, -1 → +1, +2, +3

+ve

+ve Rate of change ⇒ Minima



-ve to +ve ⇒ Minima

+ve to -ve ⇒ Maxima

Double Derivative

⇒ second derivative at all those critical points

$f(x) \rightarrow f'(x) = 0 \Rightarrow x$

$f''(x) > 0 \Rightarrow$ minima

$f''(x) < 0 \Rightarrow$ maxima

$f''(x) = 0 \Rightarrow$ Saddle Point ✗

ML

$f(x) = x^2$

$f'(x) = 2x \Rightarrow f'(x) = 0$

$2x = 0$

$x = 0$

↓

minima or maxima

$f''(x^2) = 2 > 0 \Rightarrow$ minima

Q6

Ques ⇒ $f(x) = (x+1)^2$

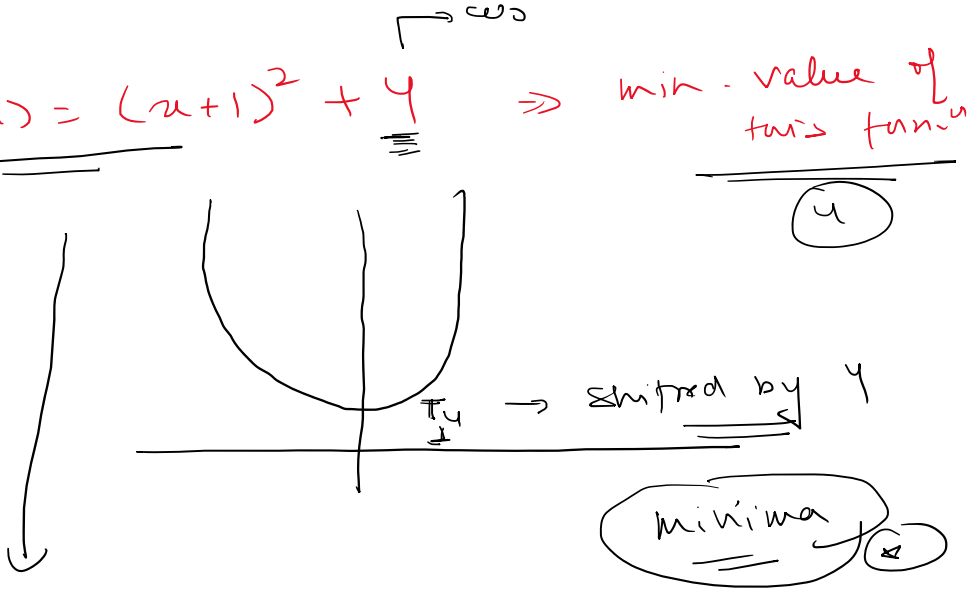
① minima ⇒ $(-1+1)^2 = 0$

formal notation

argmin w, w_0

② $x^* = \underset{x}{\text{argmin}}\ f(x)$

③ $f(x) = (x+1)^2 + y$ ⇒ min. value of this function



④ $f'(x) = 2(x+1) = 0$

$2x + 2 = 0$

$\Rightarrow x = -1$

at $x = -1$

slope $= 0$

slope $= 0$ ⇒ -1

⑤ Gradient descent ⇒

⑥ ⇒

$(x+1)^2 + 4$ ⇒ visualize ✓

$\log x^4 e^{x^2 \sin x + x^3}$ ⇒ visualize ✗

→ derivative ⇒ ✗

↓

gradient Descent

=

I V → O

= MULTIVARIATE calculus ⇒

only 1 i/p ⇒ 2 or more ivs

$z = f(x, y) = x^2 + y^2$ ⇒ 3D

↳ multiple i/p

input = $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$

i/p → vector ⇒

o/p → scalar

$(1, 2) \longrightarrow 1^2 + 2^2 = 5$ ⇒ scalar value

ML / DL ⇒ IVs

i/p → vector of all IVs

o/p → scalar no'

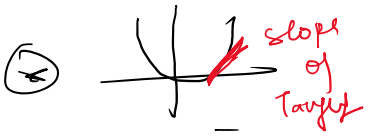
multivariate ⇒ Multivariate calculus

func^n → multiple i/p's

↳ multiple derivatives

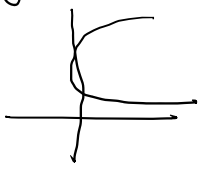
Partial Derivatives

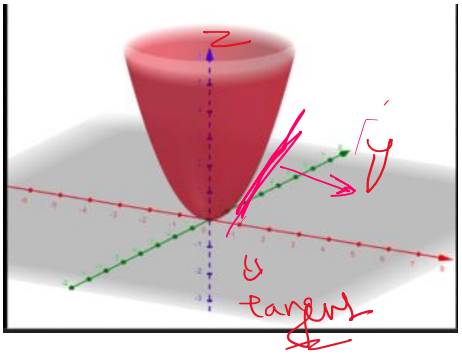
$f(x) = x^2$



$f(x) = x^2 + y^2$

$f(y) = y^2$





⇒ $\frac{df}{dx}$ ⇒ NO

$x^2 + y^2$

wrt x = ↳ (y) x

① derivative → ↑ ⇒ how func^n value is changing

↳ impact on func^n

x↑

(y) → constant

(y) → (x) constant

n diw' x, y, z, a → constant

(x) 5

$f(x)$ $x^2 + y^2 =$ (x) → y const.

(y) → (x) const.

derivative of $f(x)$ wrt. x & keeping y constant

⇓

(*) Partial Derivatives

y → x constant

$$f(x) = x^2 + y^2$$

del

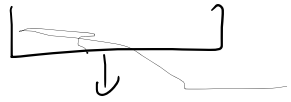
(x) $\frac{\partial f(x,y)}{\partial x} = 2x + 0 = 2x$

(y) $\frac{\partial f(x,y)}{\partial y} = 0 + 2y = 2y$

(*) loos of variables → partial derivative

gradient → $\nabla_{x,y} f(x,y) = \begin{bmatrix} \frac{\partial f(x,y)}{\partial x} \\ \vdots \end{bmatrix}$

$$\nabla_{x,y} f(x,y) = \begin{bmatrix} \frac{\partial x}{\partial f(x,y)} \\ \frac{\partial y}{\partial y} \end{bmatrix}$$



⇒ Vector Notation

func^n → $f(w_1, w_2, w_3) = w_1 x_1 + w_2 x_2 + w_3$

$$\nabla_{\vec{w}, w_3} f = \begin{bmatrix} \frac{\partial f}{\partial w_1} \\ \frac{\partial f}{\partial w_2} \\ \frac{\partial f}{\partial w_3} \end{bmatrix}$$

→ partial derivative

$\frac{df}{dx}$ → partial derivative

$w = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$

IV → 3 vars

↳ w_1 w_2 w_3

$$\nabla_{\vec{w}, w_3} f = \begin{bmatrix} \frac{\partial f}{\partial w_1} \\ \frac{\partial f}{\partial w_2} \\ \frac{\partial f}{\partial w_3} \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ 1 \end{bmatrix}$$

$\frac{\partial}{\partial w_1}(w_1 x_1 + w_2 x_2 + w_0)$

$w_1 x_1$ → wrt w_1 ⇒ x_1

$w_2 x_2$ → const ⇒ 0

w_0 → const ⇒ 0

$x_1 + x_2 + 0$

$\frac{\partial}{\partial w_2}$ ⇒ x_2

$\frac{\partial}{\partial w_3}$ ⇒ 1

5 mins Break 10:25

$x_1, x_2, 1$ → Point

→ Direction

gradient vector of f wrt w_3

$\nabla_{w_1, w_2} f$ notation

$\{w_1, w_2\}$

$f(x)$

x → keeping others as constant

Single vars → Normal derivatives

Multiple → Partial Derivatives

gradients

x → constant

$f(x, y) = 2x^2 + 4xy$ (const. → const.)

$\frac{\partial f}{\partial x} = 4x + 4y = 4x + 4y$

$\frac{\partial f}{\partial y} = 0 + 4x = 4x$

→ Chain Rule

$f(x, y) = 3 \log(xy) + 4y^2 x^3$ (f_1, f_2)

$\frac{\partial}{\partial x} \log(x) = \frac{1}{x}$

chain rule

$\frac{\partial f}{\partial x} = 3 \frac{\partial (\log xy)}{\partial xy} \cdot \frac{\partial xy}{\partial x} + 4y^2 \frac{\partial x^3}{\partial x}$

$$= 3 \times \frac{1}{xy} \times y + 4y^2 \times 3x^2$$

$$= \frac{3}{x} + 12y^2x^2$$

$$\boxed{\frac{df}{dy} = \frac{3}{y} + 8y^3}$$

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duration: 45
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Question

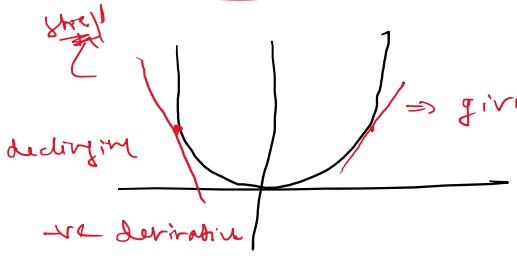
A multivariable function that takes multiple variables as input would give us multiple outputs

Choices

- [ ] True
- [x] False

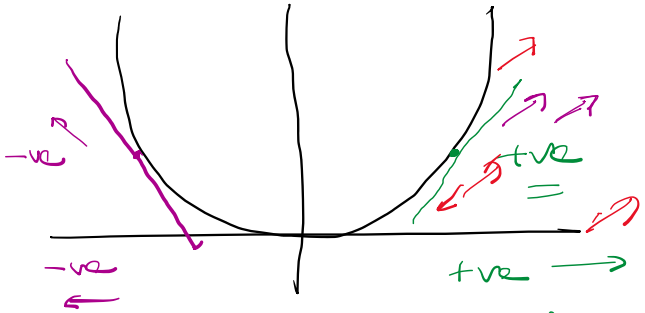
Simple func^n ⇒

$$f(x) = x^2$$



⇒ giving some dir^n ⇒ steep

(*) Multiple vars → multiple directions



func^n value will ↑

func^n will ascend →

at any point →

derivative → partial derivative

→ func^n will ↑

Loss

ascend ↑

gradient ascend ✗

① ML terms ⇒

↑ Loss or ↓ Loss

② ↓ Loss

gain func^n

ascending

max value

Loss func^n → minimize

giving dir^n of ascend

gradient → giving dir^n of ascend

we want to use gradient descent

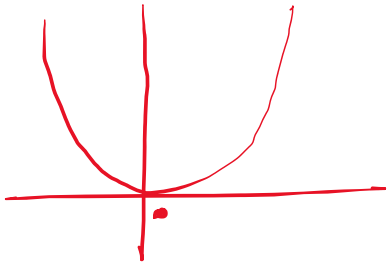
↓

go down

⇒ dir^n

(+ve) or (−ve)

gradient → dir^n of steepest ascend

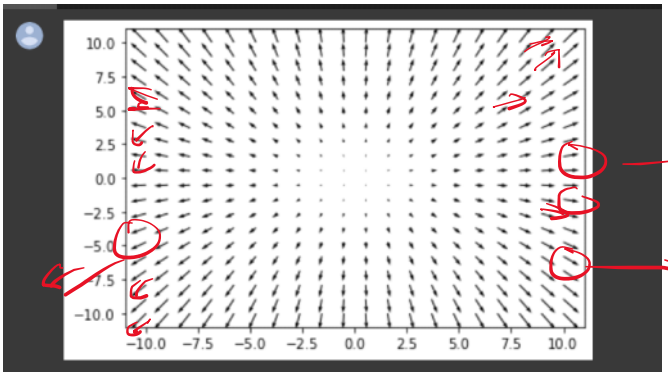


we dont want to go

⊖ Because of loss func^n

we want to ↓

⊕ gradient Descent



gradient ascending

dir^n

Large length

slope steepness

⊕ Rate of change = Higher

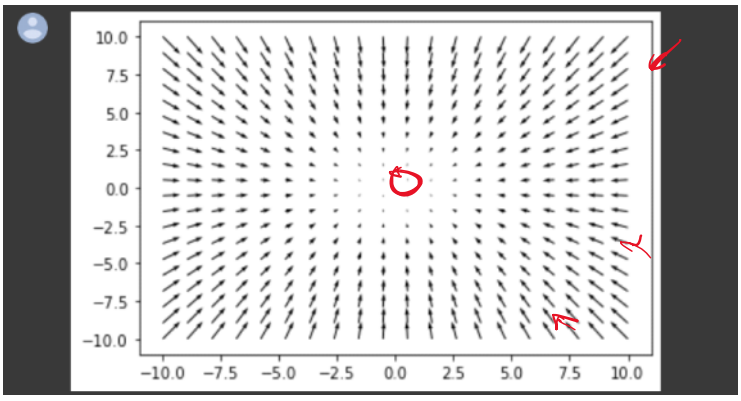
⊖ Loss → ↓

Loss → ↓

minimise loss

Best value

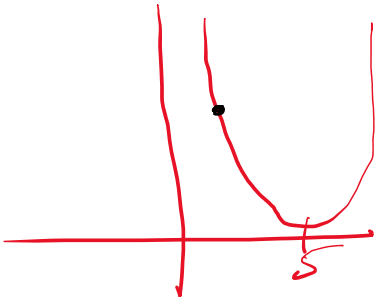
Rate of change = Higher



Loss

Gradient Descent ⇒

$$f(x) = (x-5)^2$$



solve for ⇒

$$x^* = \arg\min_x (x-5)^2$$

⇒ min.

↓

can be solved using GD

=

→ searching problem

⇒ (4) ⇒

(1) Try with Random search ⇒

(1) Random value → see how much is the loss → store

(2) Random value → Loss

Loss value ↓ → keep that value of x

again search for (3)

(4) → pick random x →
→ check if loss ↓ → store ⇒

Random value ⇒ Hectic approach ⇒

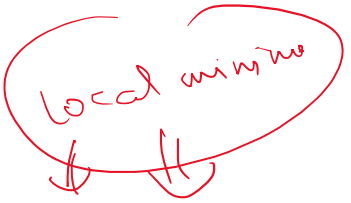
→ (turns) Bad sol^n

(1) gradient Descent ⇒

3 steps

(1) Randomly initialize variable (x)

(2) Repeat until convergence

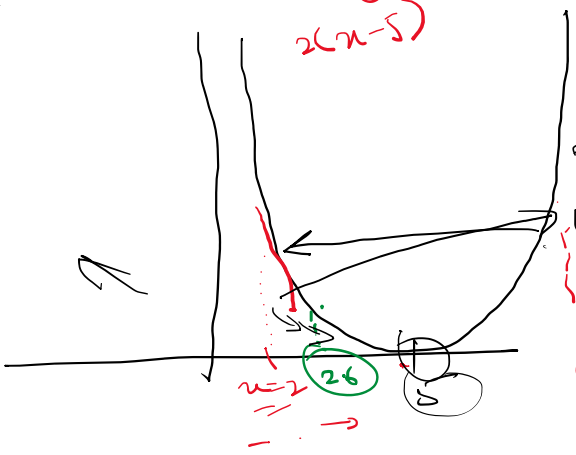


(2) Repeat until

$$x = x - \eta \frac{df}{dx}$$ → general form

$(x-5)^2$ $\quad (x-5)^2$ at $x=2$ $\quad 2(x-5)$

$x = 2$



at $x=2$ → −ve derivative

Right

& → opp. to dir^n of gradient

Learning Rate → step size → small

$x = 2 - \eta(-6)$ → units

$x = 2 + \eta(6)$

→ Right side

$\eta = 1$

$\eta = 0.1$

6 steps → 8 is too far

we should go little steps

$$x = x - \eta \frac{df}{dx}$$

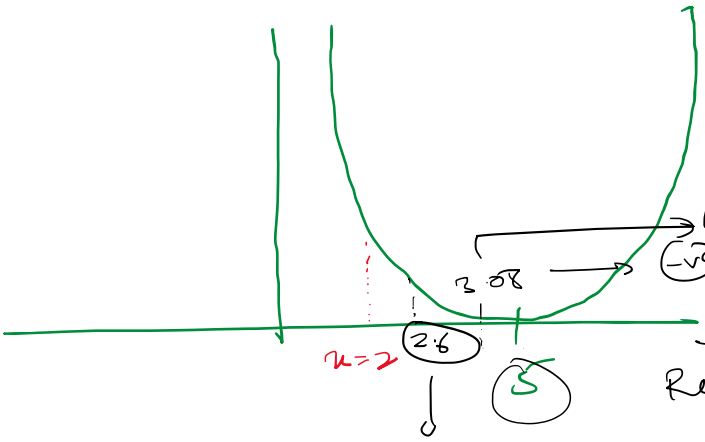
$\eta = 0.1$

$x_{updated} = 2 + 0.1 \times 6$

$= 2.6$

global minima

Overshooting problem



Repeating iteration

converge at 5

2.6 ⇒ -ve ⇒ opposite to dir^n of grad.

again → Run

$$x = x - \eta \frac{\partial f}{\partial x}$$

$$x_{updated} = 2.6 - (0.1)(-4.8)$$

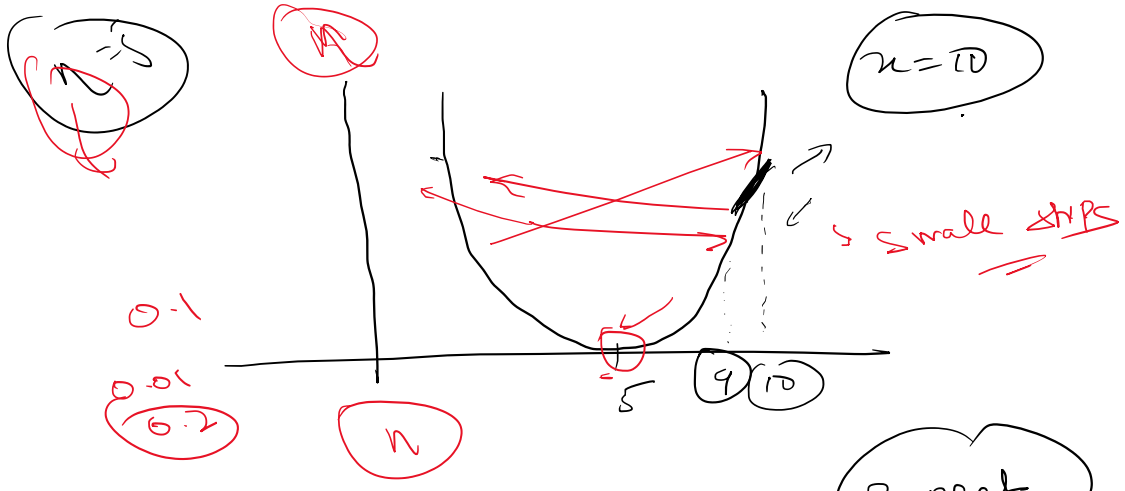
$$= 2.6 + 0.48 = 3.08$$

why will it converge at $x=5$

derivative at $x=5$ becomes 0

⇒ GD is

you can run this for any loss func^n where there is minima



(*) $x = x - \eta \frac{df}{dx}$ ⇒ Repeat

$x = 10 - (0.1)(10)$ (10×5)

$= 10 - 1 = 9$ 10 - 50

⇒ boost session

η → any value

m_t → hyperparam

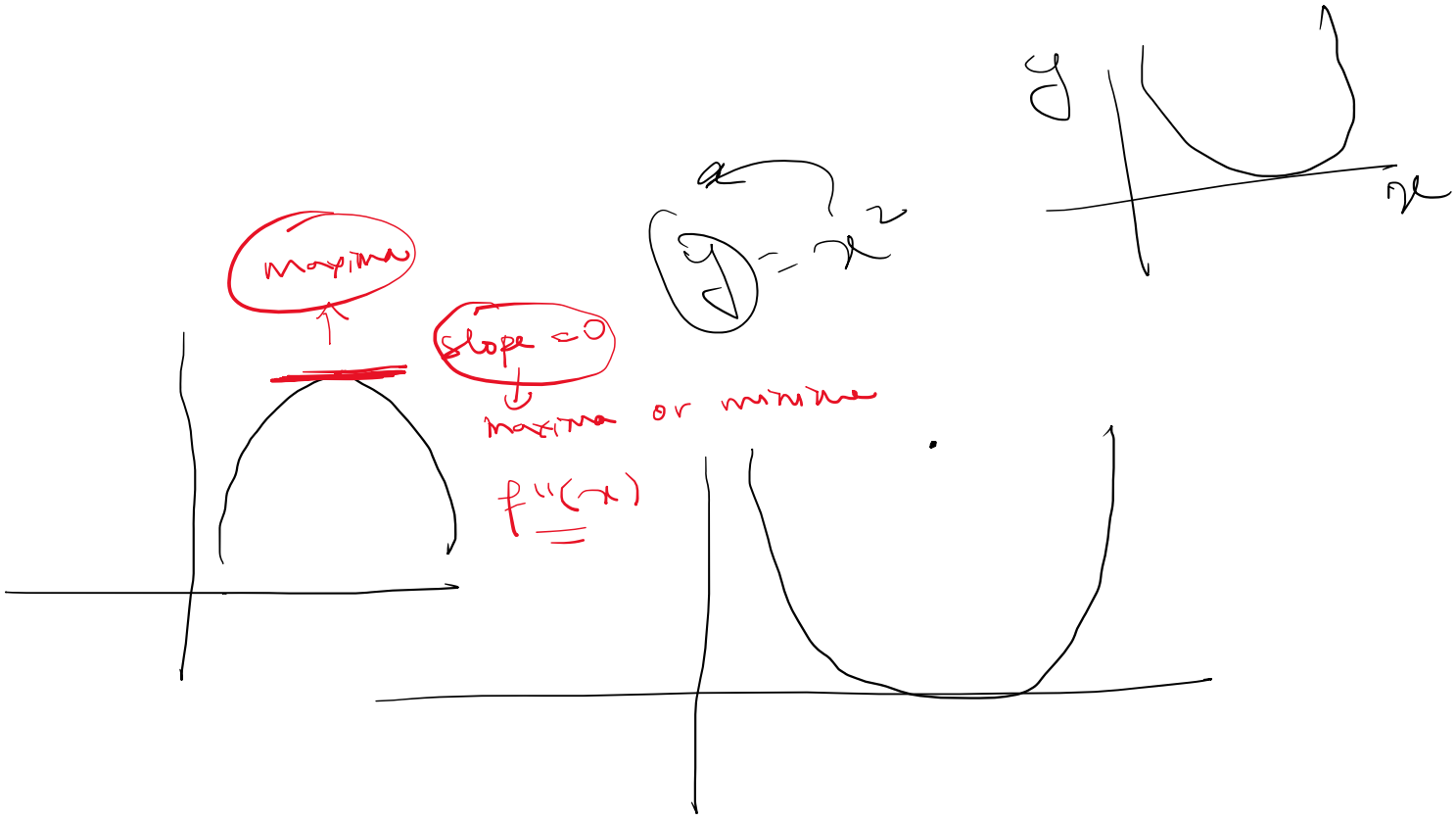
0.01

$w_0 = w_0 - \eta \cdot \frac{dL}{dw_0}$

$\vec{w} = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix}$ w_0

loss fn

$\vec{w} = \vec{w} - \eta \cdot \frac{dL}{dw}$


maxima
slope = 0
maxima or minima
f''(x) =
y
x
y = -x²